Statistics 280 — Practice Midterm 1B — Solutions

Point values per part match the exam. Normal-table values follow the table at the back of the exam; answers within rounding of these are fine.

Ex. 1 (30 points)

a (7 points)

No. This is an observational study: the scholarship is determined by the exam score, so recipients differ from non-recipients in many ways. Confounders: academic ability/preparation, motivation, family income, parents' education, quality of high school. Each one is **associated with the treatment** (higher values make scoring 1200+ more likely) and **affects the response** (graduation) on its own. The 31-point gap mixes the scholarship effect with these differences, biased in favor of the scholarship.

b (8 points)

A **regression discontinuity design**. Students just below and just above an arbitrary cutoff are very similar; whether a student lands at 1195 or 1205 is mostly luck on test day, so assignment is close to random. Estimate: 63% − 58% = **5 percentage points**. Check that the two groups are similar on things we can measure (income, high school GPA, school type), that students cannot precisely control where they land around 1200, and that nothing else changes at 1200 (e.g. another program with the same cutoff).

c (7 points)

The cutoff is no longer arbitrary for those students: the most motivated near-misses move above 1200. The just-above group gains motivated students and the just-below group loses them, so motivation becomes a confounder again. The estimate would be biased **in favor of** the scholarship (too large).

d (8 points)

Number of cells $= 4 \times 2 \times 2 \times 3 = 48$. Each needs at least 40 students, so at least $48 \times 40 =$ **1,920**. More data solves the first problem with cross tabulation (running out of data). The second problem remains: you may have **left out an important confounder** (motivation, say), and no amount of data fixes that.

Ex. 2 (30 points)

a (6 points)

Weighted average: $0.2(80) + 0.3(70) + 0.5(76) = \mathbf{75}$. The TA gets $(80 + 70 + 76)/3 = 75.33$. That gives each section equal weight even though the sections have different sizes.

b (8 points)

In each section, the mean of the squares is $SD^2 + \text{mean}^2$: $25 + 6400 = 6425$, $100 + 4900 =$

5000, $36 + 5776 = 5812$. Overall mean of squares: $0.2(6425) + 0.3(5000) + 0.5(5812) = 5691$. Variance $= 5691 - 75^2 = 66$, so SD $= \sqrt{66} \approx$ **8.12**.
It's larger than every section SD because the combined spread includes the differences *between* the section means (80, 70, 76), not just the spread within sections.

c (10 points)
Sorted: 1, 3, 4, 7, 7, 7, 9, 12, 15, 20 (each number is 10%).
25th: need at least 2.5, so 3 numbers: **4**. Median: need 5: **7**. 75th: need 7.5, so 8: **12**. IQR $= 12 - 4 =$ **8**. 90th: need 9: **15**. 91st: need 9.1, so all 10: **20**.
MAD: absolute deviations from 7: 6, 4, 3, 0, 0, 0, 2, 5, 8, 13. Sorted 0, 0, 0, 2, 3, 4, 5, 6, 8, 13. Median (5th) = **3**.

d (6 points)
The new list has 11 numbers; its median is the 6th smallest (need at least 5.5). The original list has 1, 3, 4 below 7 and three 7s. If $a \leq 7$, the four smallest are a, 1, 3, 4 in some order and positions 5–7 are 7s. If $a > 7$, positions 4–6 are 7s. Either way the median is 7. So we need $(85 + a)/11 = 7$, giving $a = -8$. Check: −8, 1, 3, 4, 7, 7, 7, 9, 12, 15, 20 has sum 77, mean 7, median 7.

Ex. 3 (35 points)

a (5 points)
Known areas: $4 \times 5 = 20\%$, $6 \times 5 = 30\%$, $1.5 \times 10 = 15\%$, $0.5 \times 20 = 10\%$: total 75%. The missing block has 25% over 5 years: height **5% per year**. $0.25 \times 400 =$ **100 people**.

b (6 points)
[22,25): $6 \times 3 = 18\%$; [25,30): 25%; [30,35): $1.5 \times 5 = 7.5\%$. Total 50.5%, so $0.505 \times 400 =$ **202 people**.

c (8 points)
Cumulative: 20% below 20, 50% below 25. Median **= 25**.
Mean $= 17.5(0.20) + 22.5(0.30) + 27.5(0.25) + 35(0.15) + 50(0.10) =$ **27.38**.

d (8 points)
Mean of squared midpoints: $17.5^2(0.2) + 22.5^2(0.3) + 27.5^2(0.25) + 35^2(0.15) + 50^2(0.1) = 835.94$. Variance $\approx 835.94 - 27.375^2 = 86.55$, SD $\approx$ **9.30**.
The true SD is **larger**: putting everyone at the midpoint throws away the spread within each block. Total spread = spread between blocks + spread within blocks.

e (8 points)
Q1: 20% below 20; need 5% of the 30% in [20,25): $20 + 5 \cdot \frac{5}{30} = 20.83$. Q3: exactly 75% below 30, so Q3 = 30. IQR ≈ 9.17.
Upper fence $= 30 + 1.5(9.17) = 43.75$. A 55-year-old is beyond it: **yes, an outlier**. Percentage above 43.75: $0.5 \times (60 - 43.75) =$ **8.13%** (about 32 people).

Lower fence $= 20.83 - 13.75 = 7.08$, below the youngest age (15). No low outliers; the lower whisker reaches the minimum.

Ex. 4 (35 points)

a (5 points)
$z = (450 - 500)/40 = -1.25$ and $(560 - 500)/40 = 1.5$. $78.87/2 + 86.64/2 = 39.44 + 43.32 =$ **82.76%**.

b (5 points)
2.5% in the left tail means 95% in the middle: $z = -1.96$. $T = 500 - 1.96(40) =$ **421.6 hours**.

c (5 points)
$z = 30/40 = 0.75$. Area within 0.75 is 54.67%, so $50 + 27.34 = 77.34$: about the **77th percentile**.

d (6 points)
5% in the left tail means 90% in the middle: $z = -1.65$. $(450 - 500)/\sigma = -1.65$, so $\sigma \approx 30.3$ **hours**.

e (5 points)
Markov: $28/70 =$ **40%**. Chebyshev: 70 is $(70 - 28)/10 = 4.2$ SDs above, so at most $1/4.2^2 =$ **5.67%**. Chebyshev is much better.

f (5 points)
Zero is $28/10 = 2.8$ SDs below the mean. Chebyshev: at most $1/2.8^2 =$ **12.76%**. (Markov only bounds the upper tail, so it doesn't help here.)

g (4 points)
No. 48 is $(48 - 28)/10 = 2$ SDs above the mean, and Chebyshev says at most $1/4 = 25\%$ of values can be 2 or more SDs from the mean. 30% is impossible.

Ex. 5 (30 points)

a (8 points)
$\overline{xy} = 79{,}550/50 = 1{,}591$. cov $= \overline{xy} - \bar{x}\bar{y} = 1591 - 40 \times 38 = 71$. $r = 71/(10 \times 9) =$ **0.79**. Months multiply x by 12, a positive constant, so r is **unchanged**.

b (10 points)
(i) Slope from (2,10) to (4,7): −1.5. From (4,7) to (8,1): −6/4 = −1.5. These three points lie on $y = 13 - 1.5x$. Put the third point on the line: $5.5 = 13 - 1.5x_3$, so $x_3 = 5$. All points on a line with negative slope means $r = -1$.
(ii) Need $\overline{xy} = \bar{x}\bar{y}$. $\bar{y} = 23.5/4 = 5.875$, $\bar{x} = (14 + x_3)/4$, $\sum xy = 20 + 28 + 5.5x_3 + 8 =$

$56 + 5.5x_3$. So $\frac{56+5.5x_3}{4} = \frac{14+x_3}{4} \cdot 5.875$, i.e. $56 + 5.5x_3 = 82.25 + 5.875x_3$, so $x_3 = -70$. One wild point can wipe out a perfect linear relationship.

c (6 points)
Plot 1: −0.7. Plot 2: 0.1 (a strong positive cloud dragged down by one outlier at the lower right). Plot 3: 0.4. Plot 4: 0.9. (Actual correlations: -0.7, 0.14, 0.4, 0.9.)

d (6 points)
(i) **Larger.** 12-year-olds are both taller and heavier, so combining extends the cloud along the same upward trend; the spread along the line grows relative to the scatter around it. (Combining increases r when the relationships are similar and the groups cover different ranges of x.)
(ii) **Smaller.** Restricting the range of x (attenuation) removes much of the spread along the line while the vertical scatter around it stays the same.

Ex. 6 (40 points)

a (8 points)
Slopes: $r\,SD_y/SD_x = 0.5$ and $r\,SD_x/SD_y = 0.5$. Their product is $r^2 = 0.25$, so $r = 0.5$ (positive, since the slopes are positive). Both lines pass through $(\bar{x}, \bar{y})$: $\bar{y} = 0.5\bar{x} + 35$ and $\bar{x} = 0.5\bar{y} + 33.5$. Substituting: $\bar{x} = 0.25\bar{x} + 51$, so $\bar{x} = 68$ and $\bar{y} = 69$.

b (6 points)
$0.5 = 0.5 \cdot SD_y/3$, so $SD_y = 3$. RMS error $= \sqrt{1 - 0.25} \times 3 = \mathbf{2.60}$.

c (8 points)
Predicted son's height: $0.5(74) + 35 = 72$. In this vertical strip, sons' heights are roughly normal with mean 72 and SD 2.60. Taller than 74: $z = 2/2.6 = 0.77$. Area within 0.77 $\approx 55.9\%$, so about $(100 - 55.9)/2 \approx \mathbf{22\%}$.

d (6 points)
This is the **regression fallacy**. SD(sons) = SD(fathers) = 3, so the spread is not shrinking. The regression effect describes *averages* in vertical strips; the scatter around the line restores the spread. It also runs backwards: fathers of tall sons are, on average, shorter than their sons. That couldn't happen if heights were really converging.

e (6 points)
25th percentile: $z = -0.67$. Predicted son's $z = 0.5(-0.67) = -0.335$. Area within 0.335 $\approx 26.2\%$, so the left tail is about $(100 - 26.2)/2 \approx$ **37th percentile**.

f (6 points)
Use the other line: $\hat{x} = 0.5(72) + 33.5 =$ **69.5 inches**. Predicting fathers from sons averages over a *horizontal* strip, and that line regresses toward the fathers' mean of 68. The two regression lines are different, so you can't just "undo" part c).